

Pavan Kumar Gannaju

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PROFESSIONAL SUMMARY

AI Security / Platform Engineer focused on Kubernetes admission control, AI governance, prompt-injection defense, secure model artifacts, backend APIs, and CI-backed verification. Built evidence-backed systems including a Kubernetes admission controller for signed images and verified model artifacts, an AI guardrail gateway, a secure desktop-agent orchestration prototype, and a cloud-validated backend ledger. Strong focus on security boundaries, adversarial testing, supply-chain verification, failure handling, and reproducible evaluation.

SKILLS

Languages: Python, Go, Java, Rust, SQL, JavaScript/TypeScript

AI Security: Prompt-injection defense, PII detection, guardrail design, adversarial testing, detector evaluation, secure agent workflows

Platform Security: Kubernetes admission control, controller-runtime, cosign, safetensors validation, SBOM, Trivy, Helm, Kustomize

Backend: FastAPI, Spring Boot, REST APIs, gRPC, JWT/OAuth2, OpenAPI

Data & Messaging: PostgreSQL, Redis, Kafka, SQLite, FAISS

Cloud & Infrastructure: Azure, Docker, Kubernetes, kind, GitHub Actions, Linux

Testing & Evaluation: envtest, fuzzing, property-based testing, k6, Testcontainers, CI, benchmark analysis

EXPERIENCE

AI Intern with Research Team Lead Responsibilities, Prodigious AI Technologies Pvt. Ltd., Delhi *Mar 2025 – Nov 2025 | Remote*

- Led a 5-intern R&D team focused on generative-AI security, trustworthy AI systems, and backend workflow components.
- Built modular Python backend workflows with API integration utilities, input-validation logic, and dedicated test scenarios for internal AI workflow systems.
- Ran empirical benchmarking, reviewed implementation decisions, coordinated sprint delivery, and contributed to research manuscripts on voice-cloning input control and defense-in-depth against prompt injection, data poisoning, and adversarial manipulation.

Gen AI / LLM Intern, Digital Nexus AI, Jersey City, NJ, USA

May 2025 – Sep 2025 | Remote, Part-Time

- Built Python/FastAPI REST APIs and backend services for service-to-service communication in AI workflow systems.
- Implemented request-processing flows, input validation, and integration utilities for AI-enabled backend services.
- Diagnosed backend defects from logs and request traces, reproduced failures, and added unit/functional tests before release handoff.

PROJECTS

ModelGate — Kubernetes Admission Control for ML Workloads | *Go · Kubernetes · controller-runtime · cosign · safetensors · Helm · Kustomize · envtest* | [GitHub](#)

- Built a Kubernetes admission webhook that admits only cosign-signed container images and hash-verified safetensors model artifacts while denying pickle-like payloads, privileged containers, host access, and unsafe workload updates.
- Found and fixed two real admission bypasses: post-admission image swaps via kubectl set image and ephemeral-container injection.
- Validated with unit tests, envtest against a real API server, fuzzing, real cosign sign/verify integration, kind-cluster smoke tests, Helm/Kustomize install paths, and Azure ACR evidence.

Gatekeeper — AI Security Guardrail Gateway | *Python · FastAPI · Redis · FAISS · PyTorch · SQLite · Docker · Trivy* | [GitHub](#)

- Built a fail-closed AI governance gateway that screens prompts for injection risk, PII, policy violations, and unsafe requests before they reach a language model.
- Combined symbolic rules, PII detection, FAISS similarity search, detector fusion, Redis-backed rate limiting, circuit breaking, and audit logging.
- Evaluated on a 6,933-prompt suite and documented operating-point tradeoffs, false positives, false negatives, profiling results, and known limitations.
- Found and fixed an authorization bypass where client-supplied role data could disable guardrails; capability is now resolved server-side.

SentinAI — Secure Desktop-Agent Orchestration Prototype | *Python · FastAPI · Windows Automation · SQLite · Ollama* | [GitHub](#)

- Led a 5-member B.Tech thesis team building a Windows desktop-agent prototype that treats the LLM as untrusted and enforces allowlists, filesystem sandboxing, keyword filtering, confirmation gates, and postcondition checks outside the model.
- Recorded 96.7% end-to-end task success across 120 evaluated tasks using independent Windows OS-state checks rather than the agent's own report.

- Blocked 66/66 adversarial cases in the recorded fuzz suite, including shell injection, path traversal, forbidden intents, and random-noise inputs.
- Scope is intentionally bounded: Windows-only, with 6 of 19 intents still lacking independent postcondition checks.

LedgerLine — Double-Entry Financial Ledger / *Java 21 · Spring Boot · PostgreSQL · Kafka · Docker · Azure · GitHub Actions · k6 · Testcontainers* / [GitHub](#)

- Built a double-entry ledger where PostgreSQL enforces zero-sum postings and non-negative balances, with idempotent transfers, deterministic lock ordering, a transactional outbox feeding Kafka projections, and JWT-secured APIs.
- Validated on Azure using ACR, Azure PostgreSQL Flexible Server, and VM-based Docker; smoke checks, k6 correctness profile, PostgreSQL invariant checks, and resource-group teardown all passed.
- Backed by CI with Testcontainers integration tests, failure-path tests, post-load invariant verification, Docker SBOMs, and Trivy scans.

EDUCATION

M.Sc. Research in Computer and Systems Engineering — Incoming Student | Technische Universität Ilmenau, Germany · April 2027

B.Tech in Computer Science and Engineering (AI/ML) — Jawaharlal Nehru Technological University Hyderabad | Graduated April 2026 | CGPA: 8.01/10.0

CERTIFICATIONS

- Google Cybersecurity Professional Certificate
- Microsoft Cybersecurity & OS Fundamentals
- SAP Code Unnati Advanced Training Program

CODING & LEADERSHIP

- Solved 530+ LeetCode problems, including 86 Hard; solved 249 Codeforces problems, including 218 in 2026.
- Led 5-member development/research teams during internship and B.Tech thesis work; organized and mentored university hackathons with 50+ participants.